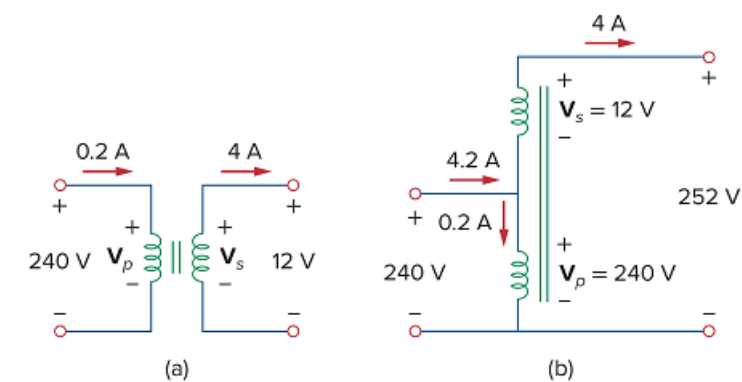


## Problem

Refer to Fig. 13.43. If the two-winding transformer is a 60-VA, 120 V/10 V transformer, what is the power rating of the autotransformer?

Figure 13.43



## Step-by-step solution

## Step 1 of 5

Calculate the value of the current  $I_2$ .

Write the expression for Complex power  $S$  supplied by the source.

$$S = V_2 I_2$$

$$60 = 10 I_2 \left[ \text{Since, } \frac{120 \text{ V}}{10 \text{ V}} \text{ transformer, } S = 60 \text{ VA} \right]$$

$$I_2 = 6$$

Therefore, the value of the current  $I_2$  is 6 A.

## Step 2 of 5

Draw the following two-winding transformer circuit diagram:

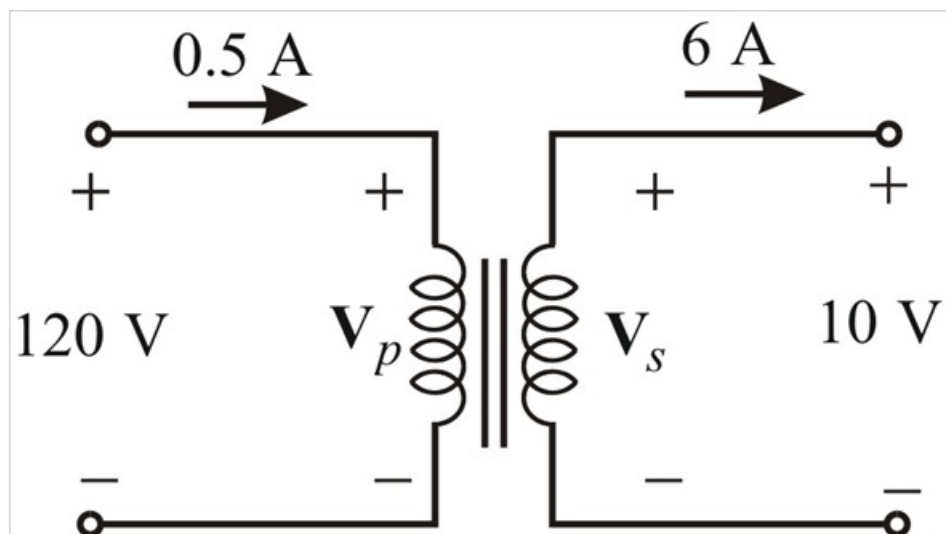


Figure 1

## Step 3 of 5

Draw the following Equivalent Auto transformer circuit diagram for Figure 1:

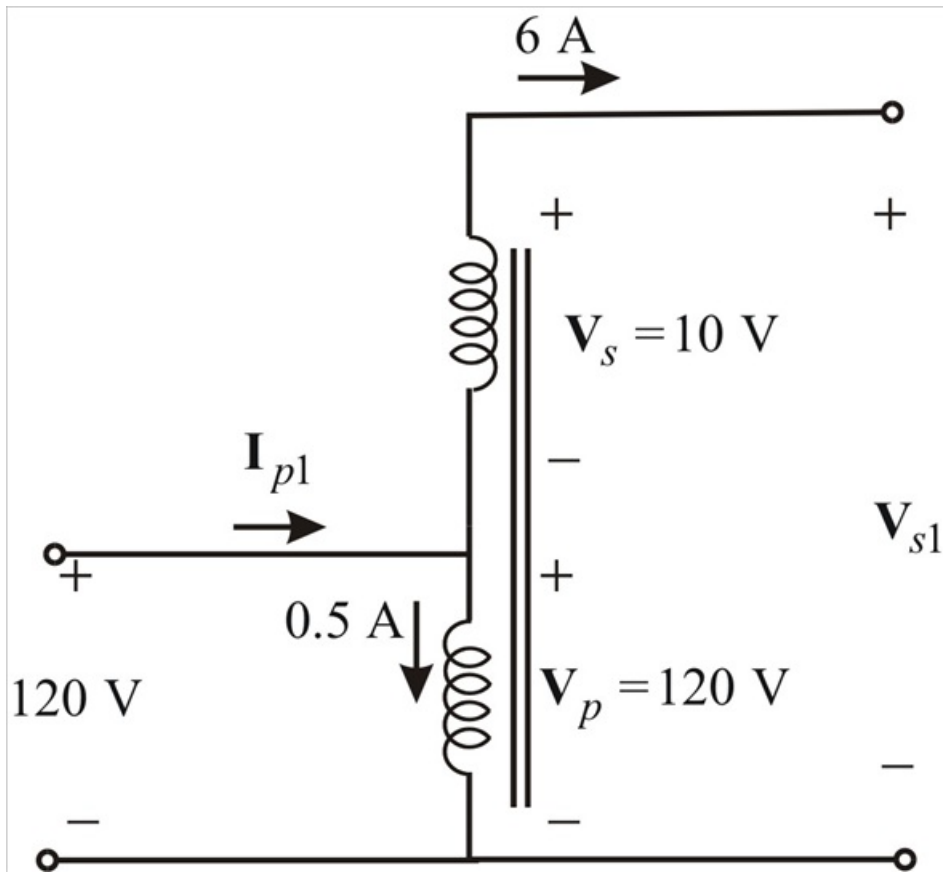


Figure 2

#### Step 4 of 5

Calculate the value of the current  $I_{p1}$ .

$$\begin{aligned} I_{p1} &= 6 + 0.5 \\ &= 6.5 \end{aligned}$$

Therefore, the value of the current  $I_{p1}$  is 6.5 A.

#### Step 5 of 5

Calculate the value of the power rating  $S_{new}$  of the auto transformer.

Write the expression for  $S_{new}$  of the auto transformer.

$$\begin{aligned} S_{new} &= V_{p1} (I_{p1}) \\ &= 120(6.5) \\ &= 780 \end{aligned}$$

Therefore, the value  $S_{new}$  of the auto transformer is 780 VA.